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import random
from collections import defaultdict

# -----
# Simulated Prompting Modes
# -----

class PromptingMode:
    ZERO_SHOT = "zero_shot"
    FEW_SHOT = "few_shot"
    CHAIN_OF_THOUGHT = "chain_of_thought"

# -----
# Simulated LLM Evaluator
# -----

class LLMSimulator:
    def __init__(self):
        # Simulated base probabilities of logical correctness
        self.accuracy_rates = {
            PromptingMode.ZERO_SHOT: 0.55,
            PromptingMode.FEW_SHOT: 0.70,
            PromptingMode.CHAIN_OF_THOUGHT: 0.82,
        }

        # Simulated hallucination rates
        self.hallucination_rates = {
            PromptingMode.ZERO_SHOT: 0.30,
            PromptingMode.FEW_SHOT: 0.18,
            PromptingMode.CHAIN_OF_THOUGHT: 0.10,
        }

        # Simulated logical fallacy occurrence rates
        self.fallacy_rates = {
            PromptingMode.ZERO_SHOT: 0.25,
            PromptingMode.FEW_SHOT: 0.15,
            PromptingMode.CHAIN_OF_THOUGHT: 0.08,
        }

    def evaluate(self, mode, trials=1000):
        results = defaultdict(int)

        for _ in range(trials):
            if random.random() < self.accuracy_rates[mode]:
                results["correct_reasoning"] += 1
            else:
                results["incorrect_reasoning"] += 1

            if random.random() < self.hallucination_rates[mode]:

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        results["hallucination"] += 1

    if random.random() < self.fallacy_rates[mode]:
        results["logical_fallacy"] += 1

    return results

# -----
# Run Experiment
# -----

def run_experiment(trials=1000):
    simulator = LLMSimulator()

    for mode in [
        PromptingMode.ZERO_SHOT,
        PromptingMode.FEW_SHOT,
        PromptingMode.CHAIN_OF_THOUGHT,
    ]:
        results = simulator.evaluate(mode, trials)
        print(f"\nMode: {mode}")
        for key, value in results.items():
            print(f"{key}: {value/trials:.2f}")

if __name__ == "__main__":
    run_experiment(trials=5000)

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Mode: zero_shot
correct_reasoning: 0.55
incorrect_reasoning: 0.45
hallucination: 0.30
logical_fallacy: 0.25

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Mode: few_shot
correct_reasoning: 0.71
logical_fallacy: 0.15
incorrect_reasoning: 0.29
hallucination: 0.18

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Mode: chain_of_thought
correct_reasoning: 0.82
logical_fallacy: 0.08
incorrect_reasoning: 0.18
hallucination: 0.10

```